

Aarya Butolia

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science Engineering, Specialization: AI & Machine Learning

Sep 2023 – Present

CGPA: 9.17 / 10

Delhi Public School, Ahmedabad

Higher Secondary Education (CBSE, Class XII)

Mar 2022 – Apr 2023

90.8%

Delhi Public School, Surat

Secondary Education (CBSE, Class X)

Mar 2021 – Apr 2023

94.6%

Technical Skills

Languages: Python (Advanced), C++, SQL, JavaScript

ML / Deep Learning: Supervised & Unsupervised Learning, LSTM, ANN, CNN, Time-Series Forecasting, Feature Engineering, Model Evaluation & Tuning

NLP & Generative AI: Bag-of-Words, TF-IDF, Word Embeddings, Transformers, LLMs, Prompt Engineering, RAG Pipelines, Vector Databases

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow / Keras, LangChain, Matplotlib, Seaborn, Streamlit, Flask, FastAPI

Data & Cloud: PostgreSQL, MySQL, Data Mining, AWS, Oracle OCI, Azure, Git / GitHub, REST APIs, YouTube Data API

Mathematics: Linear Algebra, Probability & Statistics, Calculus, Numerical Methods, Discrete Mathematics

Projects

RAG-Powered YouTube Chatbot

Apr – May 2026

Python, LangChain, RAG, Vector Database, LLM, YouTube Data API | [GitHub](#)

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline that ingests YouTube video transcripts, chunks and embeds them into a vector database, and enables natural-language Q&A grounded in video content.
- Integrated the YouTube Data API for automated transcript extraction; engineered prompt templates to reduce hallucination and improve factual grounding over long-form video corpora.
- Achieved sub-second retrieval latency via semantic similarity search; designed a conversational memory module to maintain multi-turn context across user sessions.

StockWise – AI-Powered Stock Forecasting Platform

Jan – Mar 2025

Python, LSTM, Flask, PostgreSQL, Twilio | [GitHub](#)

- Designed a deep LSTM neural network for Nifty 50 equities producing 6-month and 5-year price forecasts; applied feature engineering on OHLCV data with multi-horizon outputs.
- Engineered a model-caching pipeline reducing inference latency from 5 minutes to 2 seconds via persisted artefacts and asynchronous background retraining.
- Built a multi-source data fallback chain (yfinance to TwelveData to NSE API) for 99%+ availability; automated stop-loss execution via Twilio SMS alerts; deployed on Railway.

Movie Recommender System – NLP Content-Based Filtering

Nov – Dec 2023

Python, Scikit-learn, NLP, Streamlit, TMDb API | [GitHub](#)

- Built an NLP recommendation engine on 5,000+ films using Bag-of-Words vectorisation and cosine similarity; pipeline includes tokenisation, stemming, and stop-word removal.
- Deployed an interactive Streamlit app with real-time TMDb API integration for dynamic poster fetching.

Achievements & Competitions

- **3rd Place Overall, ExpertHire Inter-VIT Hackathon (2026)** – Built a Command Dashboard (Python + Flask) for real-time system monitoring and task orchestration, outperforming 500+ teams.
- **LeetCode** – Rating 1400+; 400+ problems solved across Arrays, Trees, Graphs, Dynamic Programming, and Binary Search (Ongoing). Also active on Codeforces and CodeChef, with focus on Dynamic Programming, Graph Algorithms, and Greedy Algorithms.

Certifications

- **Oracle OCI 2025 Generative AI Professional (96%)** – LLMs, Prompt Engineering, RAG, Generative AI on OCI (Jul 2025).
- **Microsoft Azure Data Fundamentals (DP-900) (955/1000)** – Data analytics, data warehousing, Azure services (Jun 2025).
- **Oracle OCI 2025 AI Foundations Associate** – Machine learning algorithms, neural networks, AI implementation (Jul 2025).
- **NPTEL Cloud Computing** – 92%, Top 1% Nationwide (May 2025).
- **NPTEL Introduction to IoT** – 92%, Top 2% Nationwide (May 2026).

Relevant Coursework

Machine Learning / AI: Fundamentals in AI & ML, Applied Machine Learning, Artificial Neural Networks, Deep Learning, Natural Language Processing, Reinforcement & Representation Learning, Computer Vision, Data Mining & Warehousing

Mathematics / Computer Science: Applied Linear Algebra, Probability & Statistics, Numerical Methods, Data Structures & Algorithms, Database Management Systems, Cloud Computing

Activities & Leadership

- **Open Source Contributor, GSSOC'25 (GirlScript Summer of Code)** – Resolved issues and shipped features via pull requests on AI & ML projects (Jul – Sep 2025).
- **Secretary, iOS Club, VIT Bhopal** (Dec 2025 – Present).